

LECTURE THREE

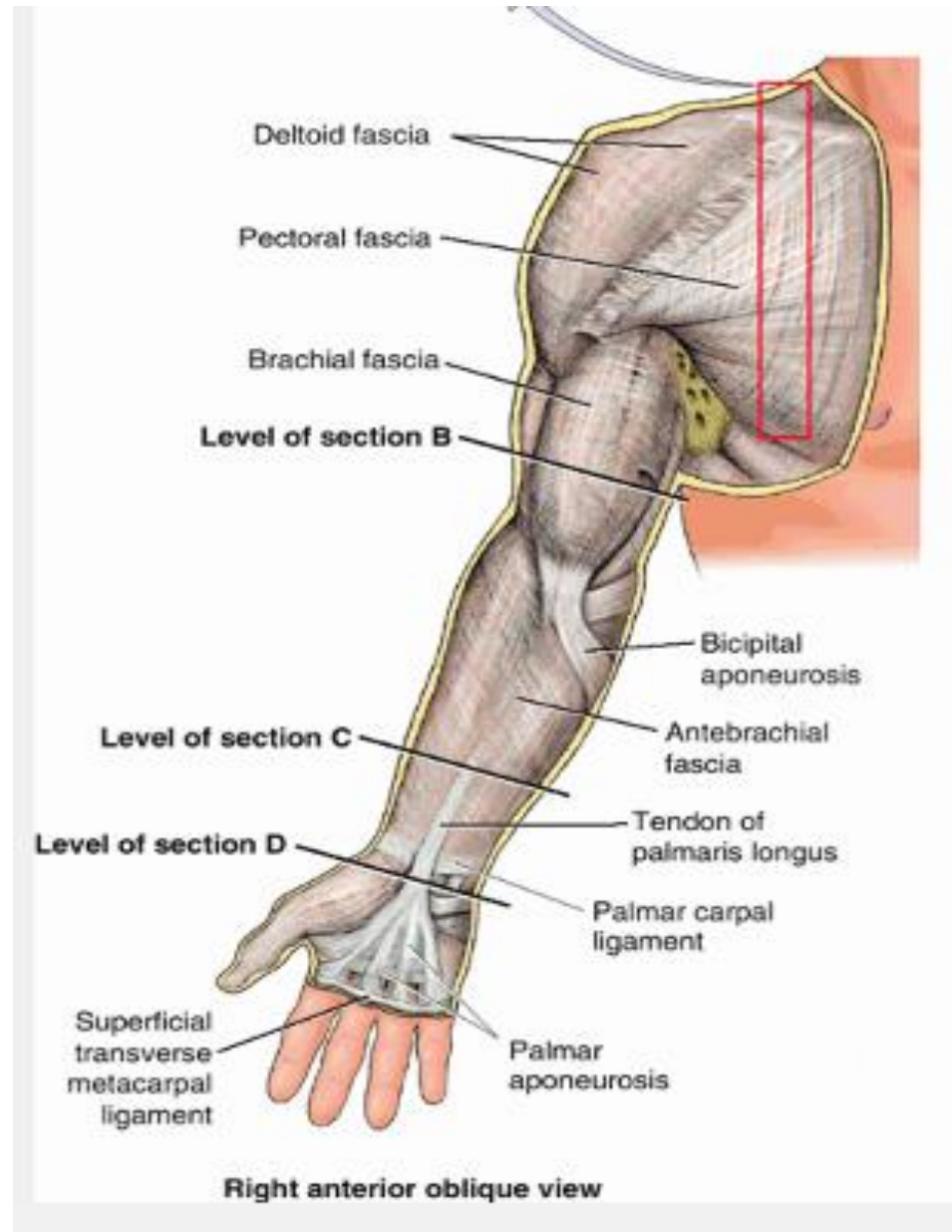
- SUPERFICIAL STRUCTURES OF UPPER LIMB

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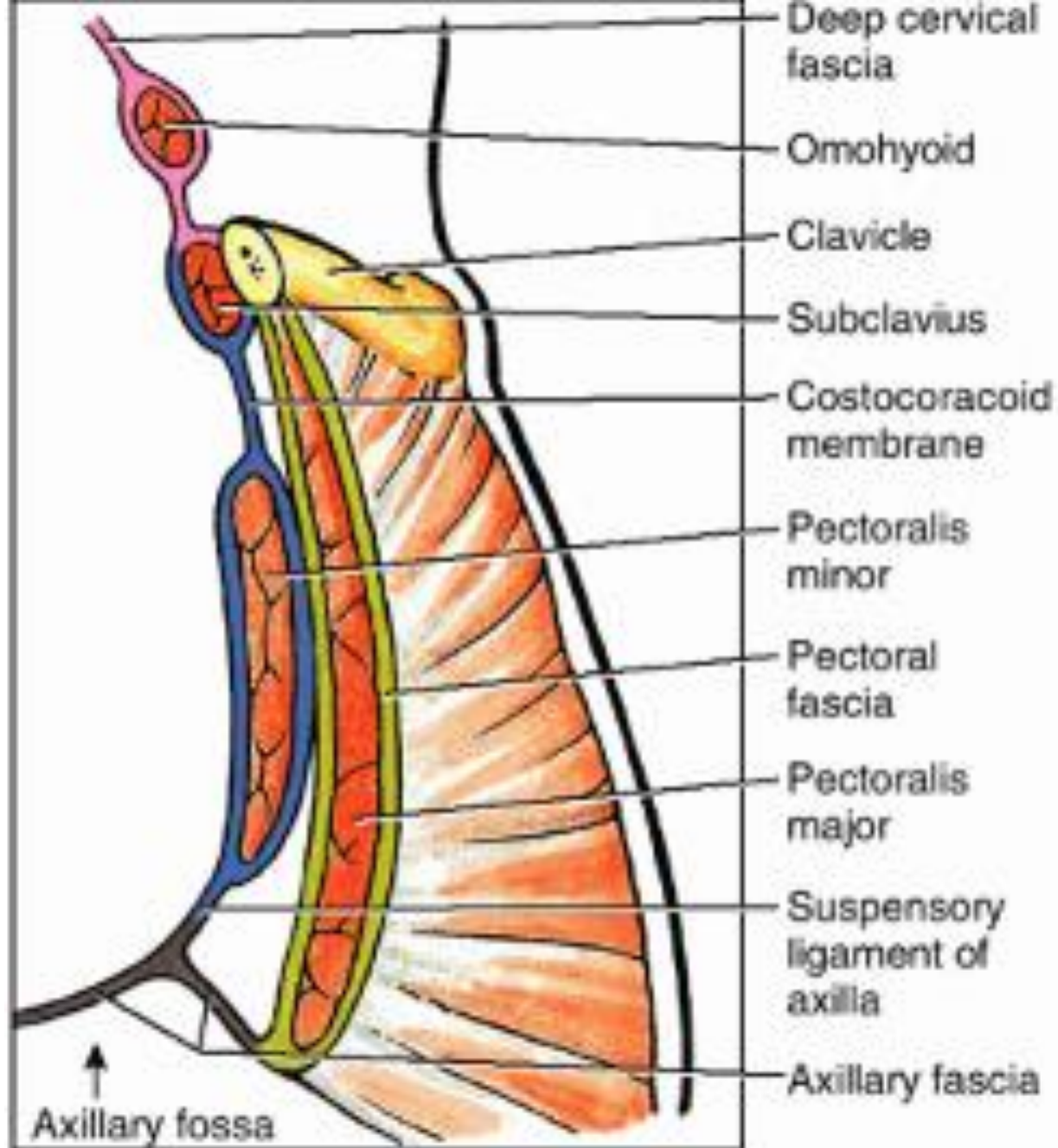
- Breast

Fascia of upper limb

- **Pectoral fascia**
 - Invests the pectoralis major
 - Continuous inferiorly with fascia of anterior abdominal wall
 - Leaves lateral border of pectoralis major and becomes **axillary fascia**



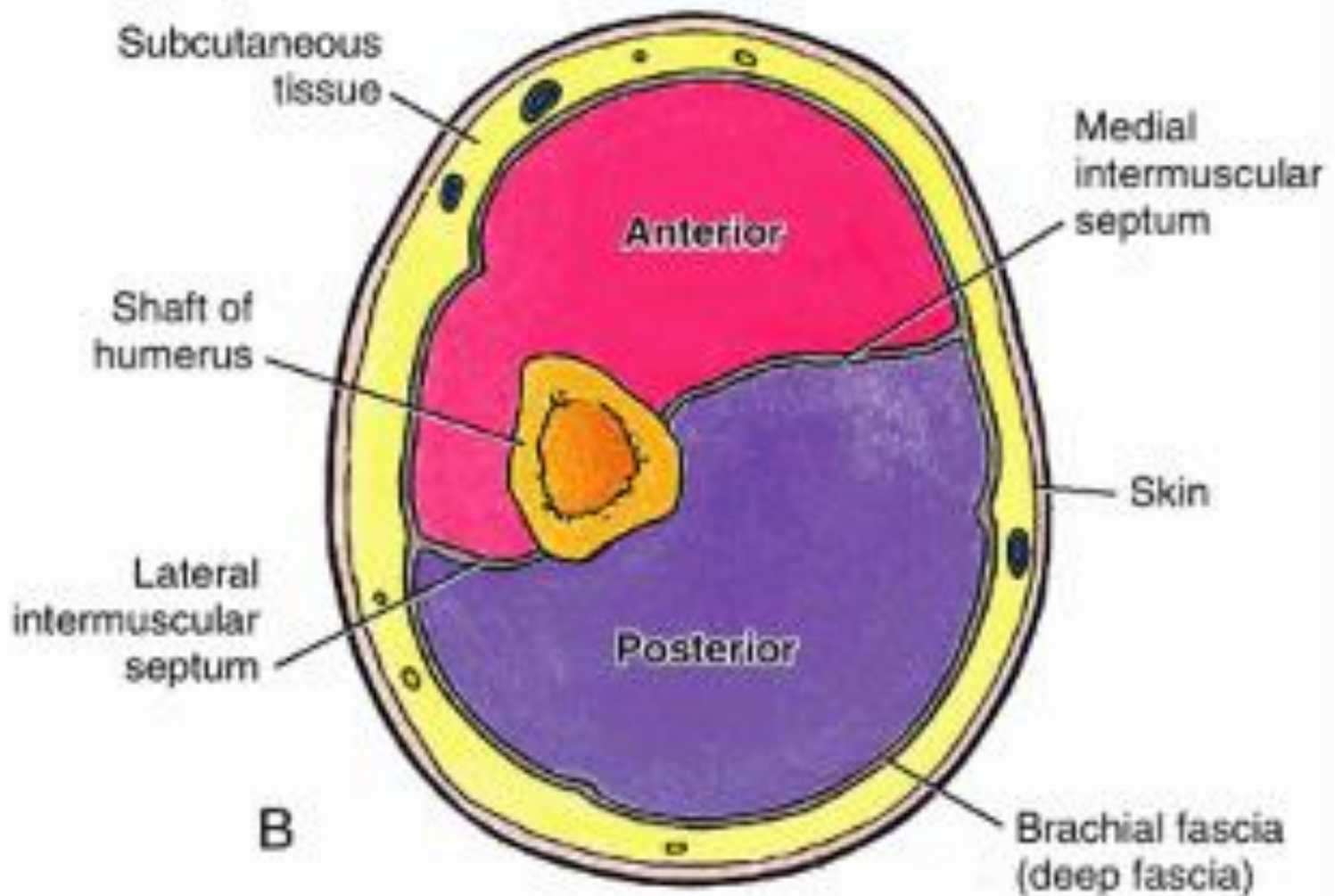
- **Clavipectoral fascia**
 - Deep to pectoralis major
 - Descends from the clavicle, enclosing subclavius and then the pectoralis minor
 - Becoming continuous inferiorly with the axillary fascia
 - **Costocoracoid membrane**
 - the part of clavipectoral fascia between pectoralis minor and subclavius
 - pierced by lateral pectoral nerve
 - **Suspensory ligament of axilla**
 - the part of the clavipectoral fascia inferior to pectoralis minor
 - supports the axillary fascia and pulls it upward and forms axillary fossa



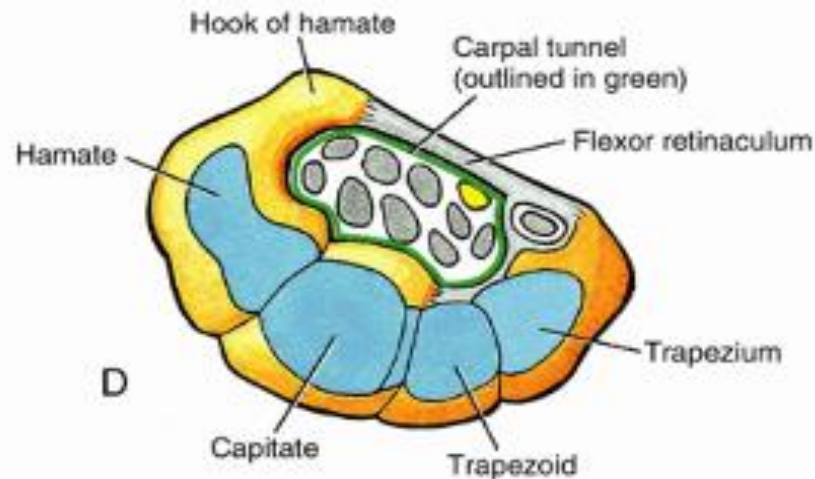
A Sagittal section

- **Deltoid fascia**
 - Invests deltoid muscle
- **Brachial fascia**
 - encloses the arm
 - continuous with deltoid, pectoral, axillary, infrapinnous and antebrachial fasciae
 - form **two intermuscular septa**
 - **medial and lateral**
 - extend from the deep surface of the brachial fascia and attach to the humerus
 - divide the arm into anterior (flexor) and posterior (extensor) fascial compartments

Fascia in A		Fascial compartments in B & C	
 Deep cervical	 Pectoral	 Anterior	
 Clavipectoral	 Axillary	 Posterior	



- **Antebrachial fascia**
 - In the forearm
 - Surrounds fascial compartments separated by interosseous membrane
 - Thickens posteriorly over the distal end to form a transverse band, **extensor retinaculum**, which holds the extensor tendons in position
 - Anteriorly at distal end forms an anterior thickenings, **palmar carpal ligament** and **flexor retinaculum**



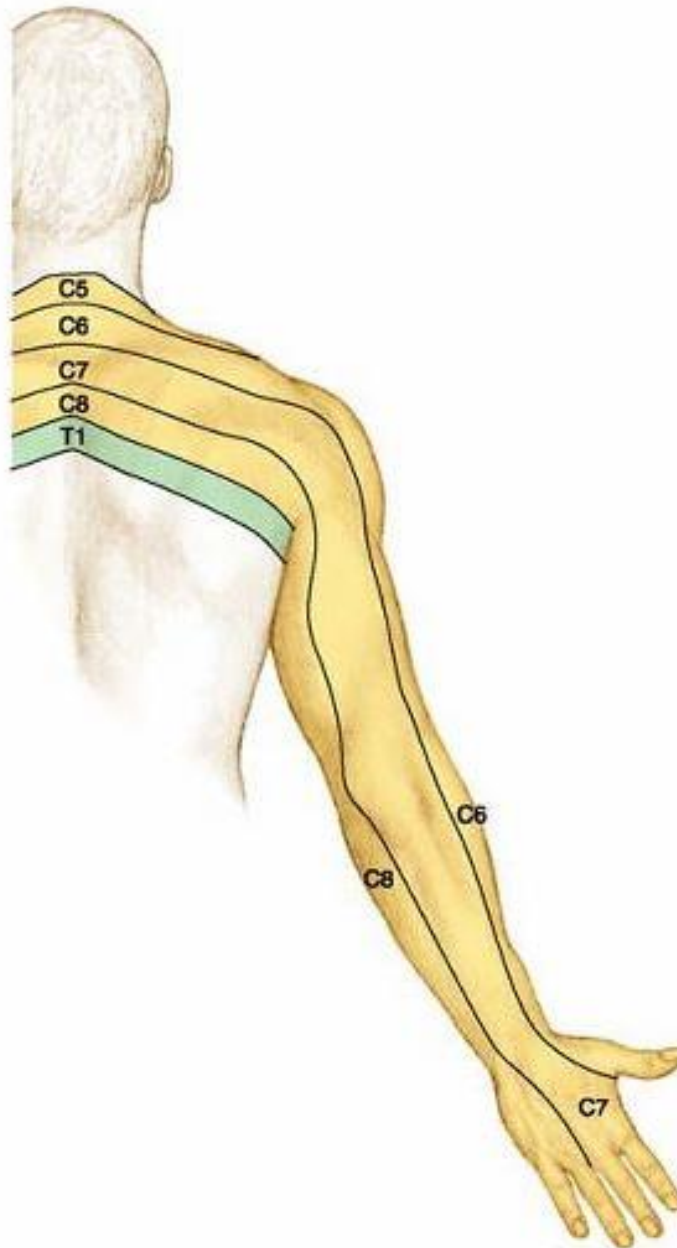
Inferior views of transverse sections

- **Palmar fascia**
 - Extension of deep fascia beyond retinacula
 - **Palmar aponeurosis**
 - Central part, thick, triangular
 - Forms four thickenings that radiate to the base of fingers
 - Superficial transverse metacarpal ligament: transverse bands distally and form the base of palmar aponeuroses
 - Ligaments extend from aponeurosis to skin, hold the skin closer

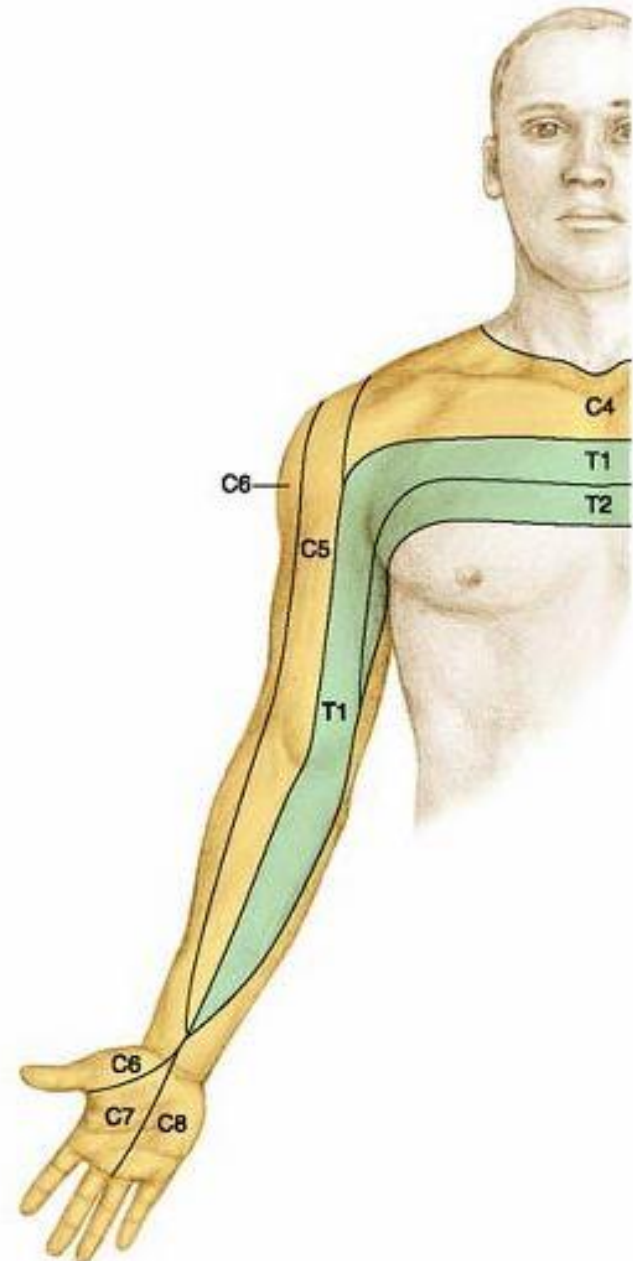
Cutaneous nerves of upper limb

- Developmentally the limbs grow as lateral protrusions of trunk, with thumb located on cranial side
- Most cutaneous nerves of upper limb are derived from brachial plexus
- Cutaneous nerves to shoulder are derived from cervical plexus
- C4: base of the neck, shoulder
- C5: arm laterally
- C6: forearm laterally and thumb
- C7: middle three fingers and middle of posterior surface
- C8: little finger, the medial side
- T1: middle of forearm to axilla
- T2: small part of arm and axilla

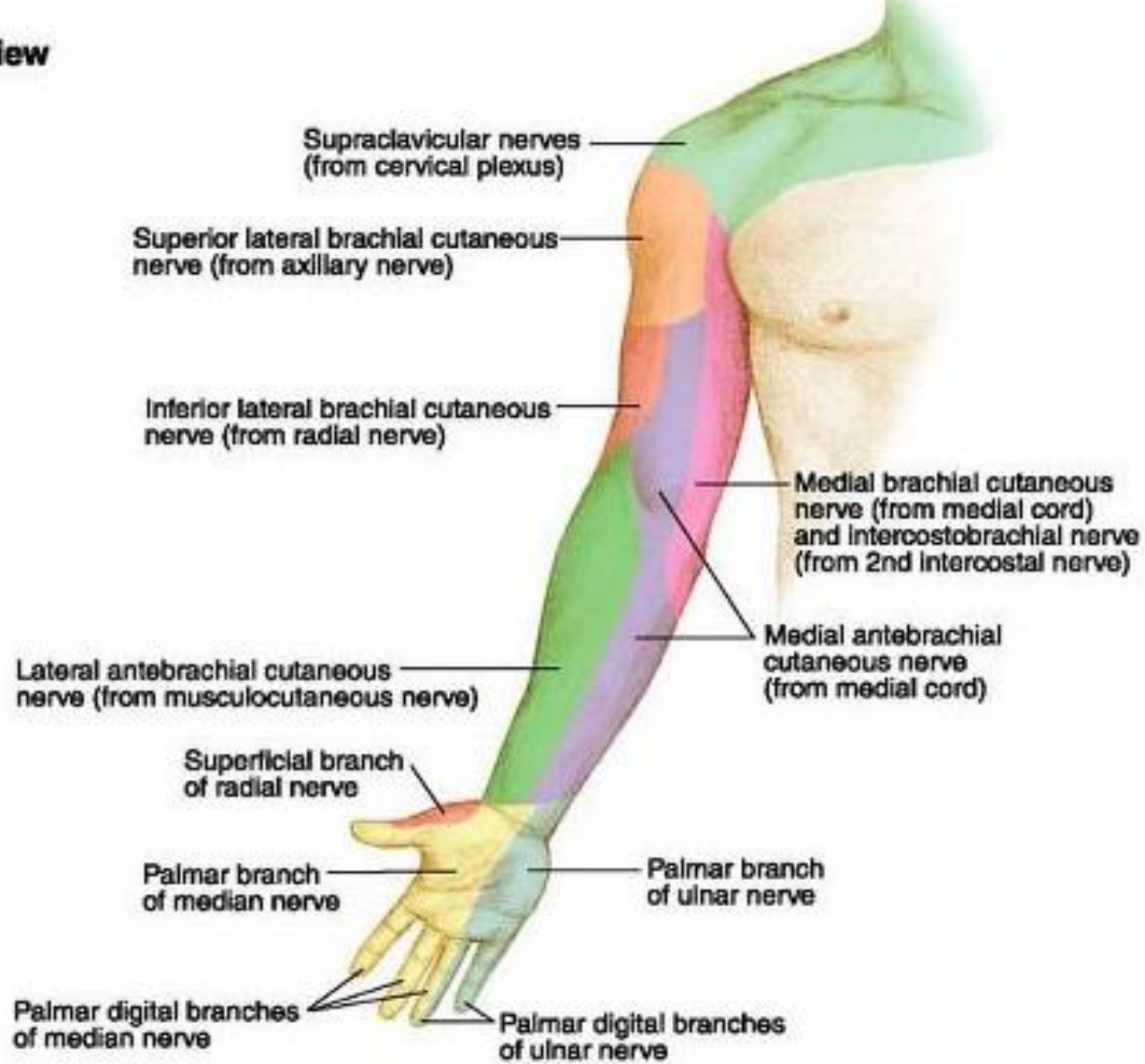
A. Posterior view



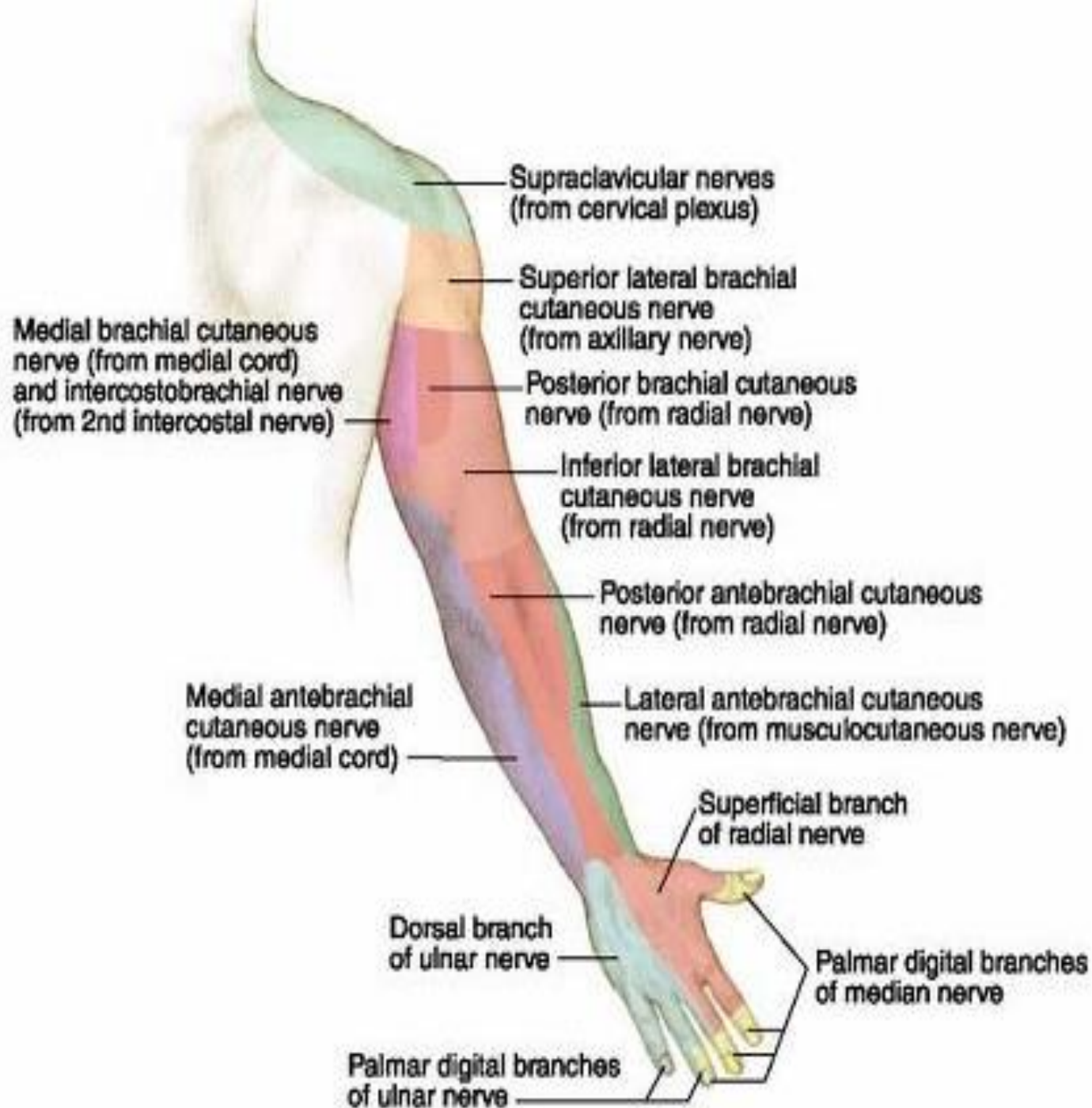
B. Anterior view

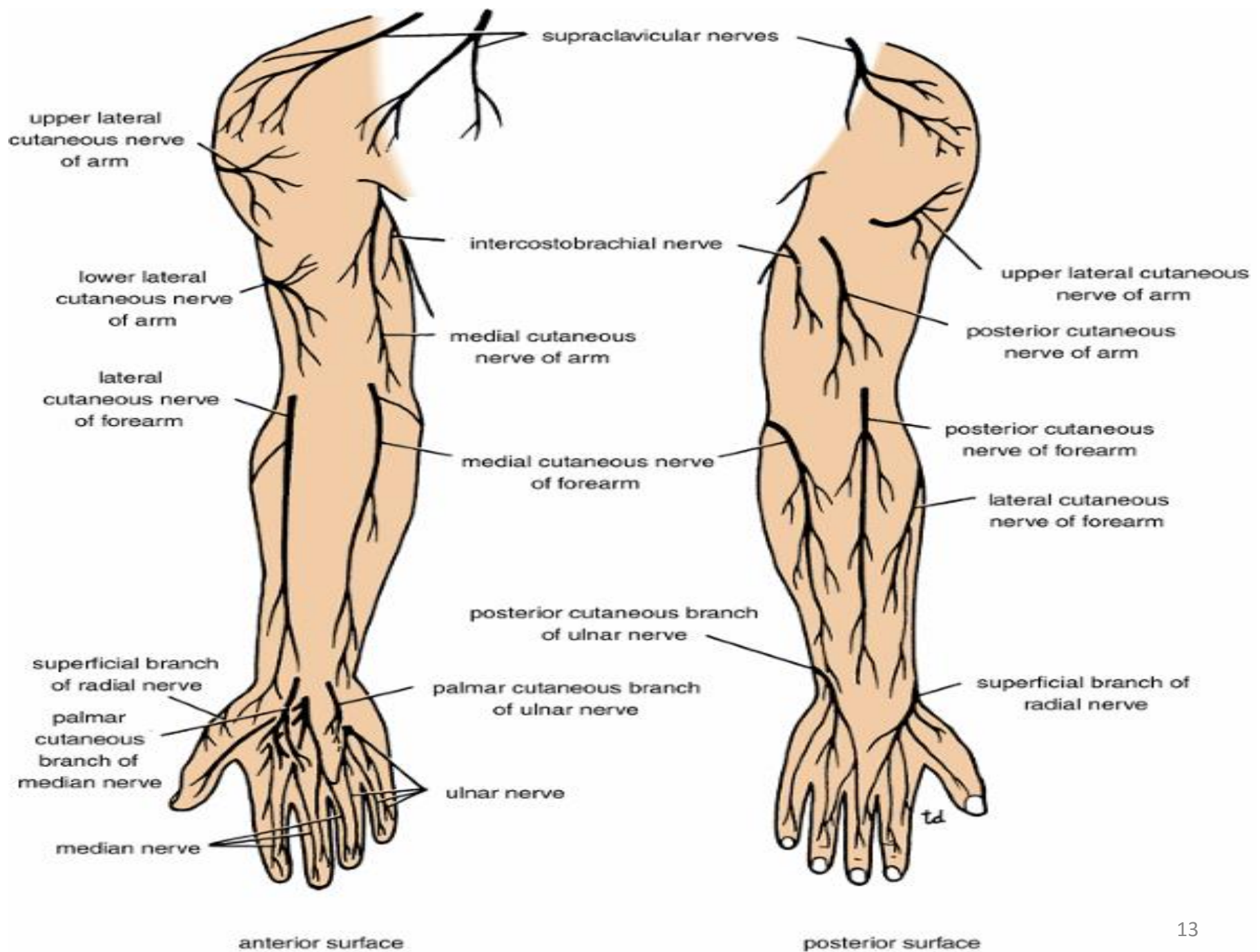


A. Anterior view

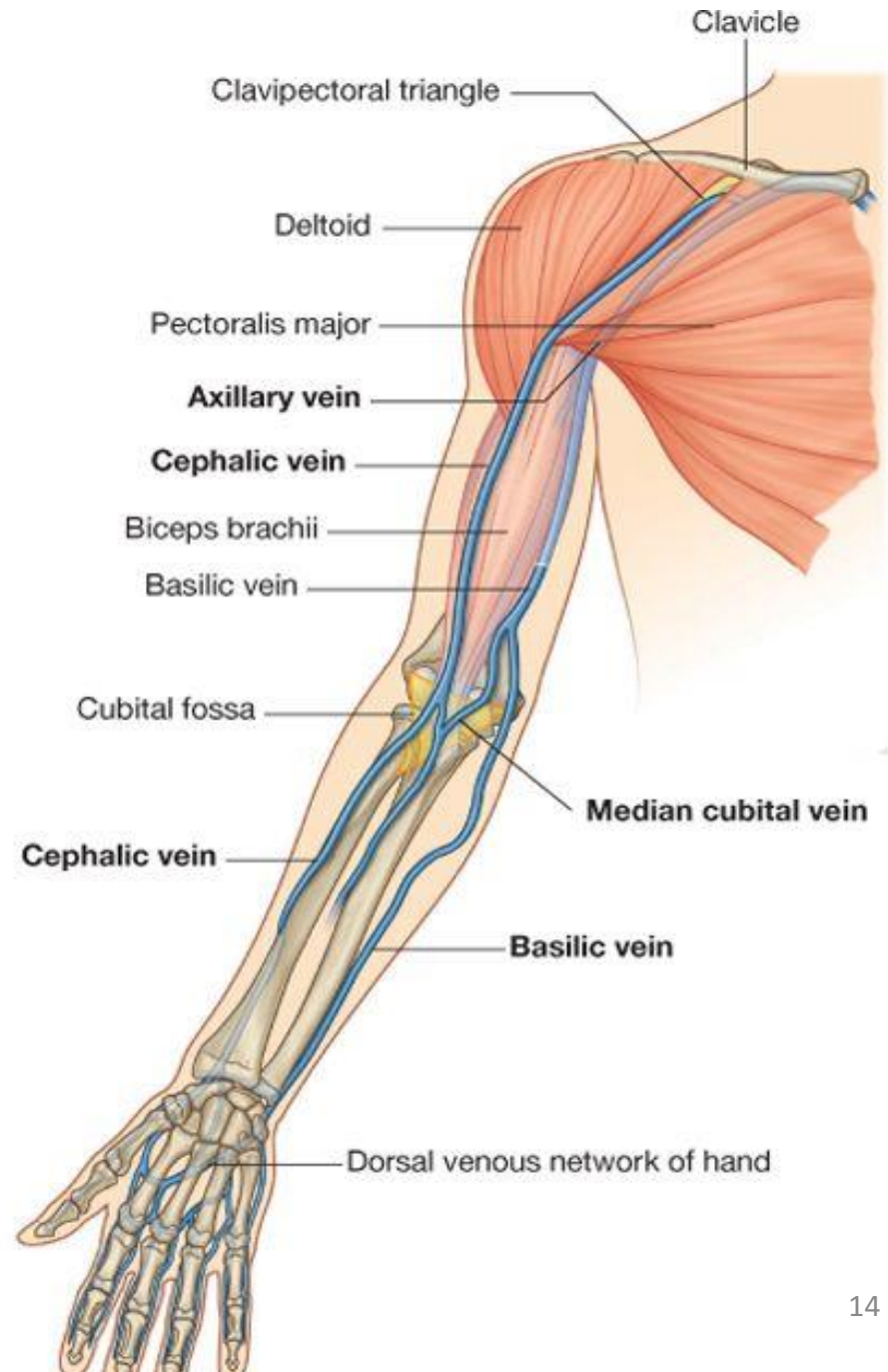
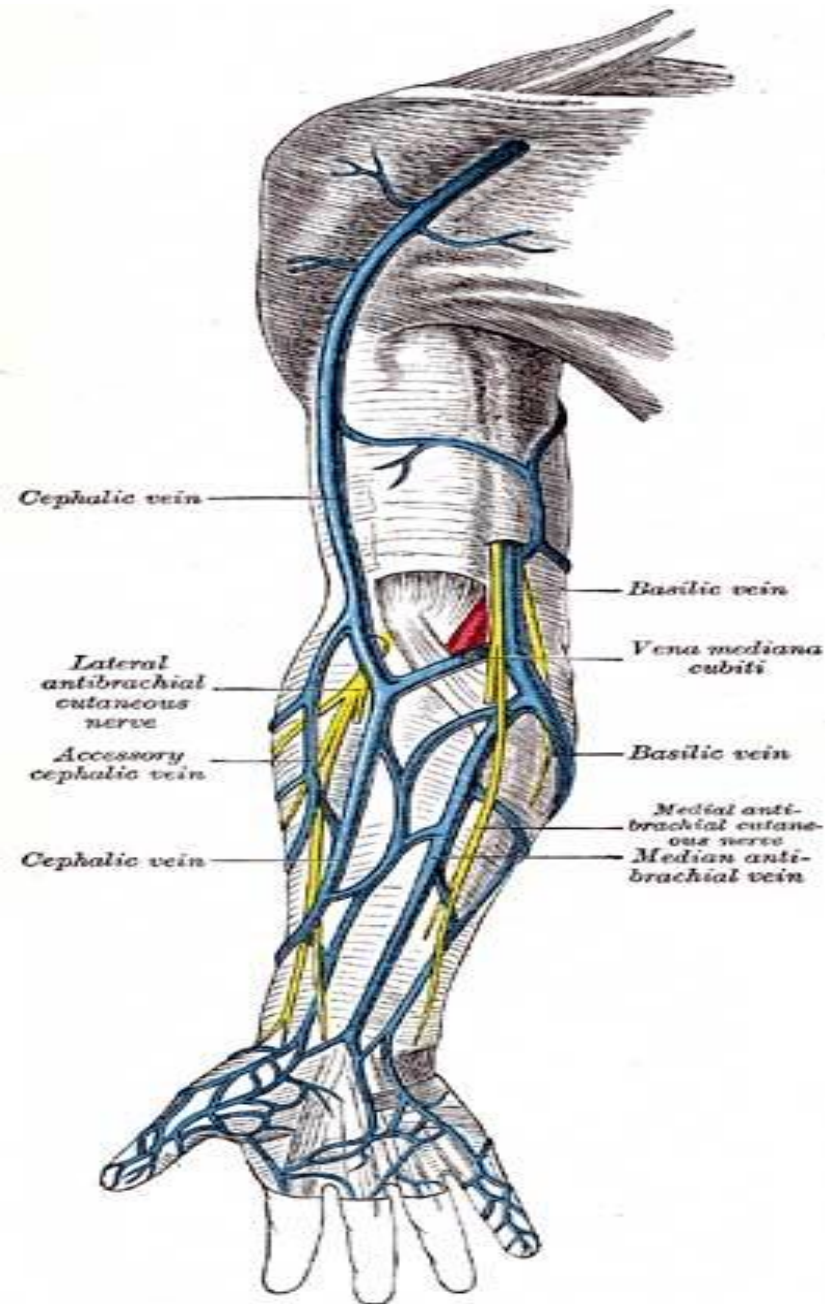


Posterior view





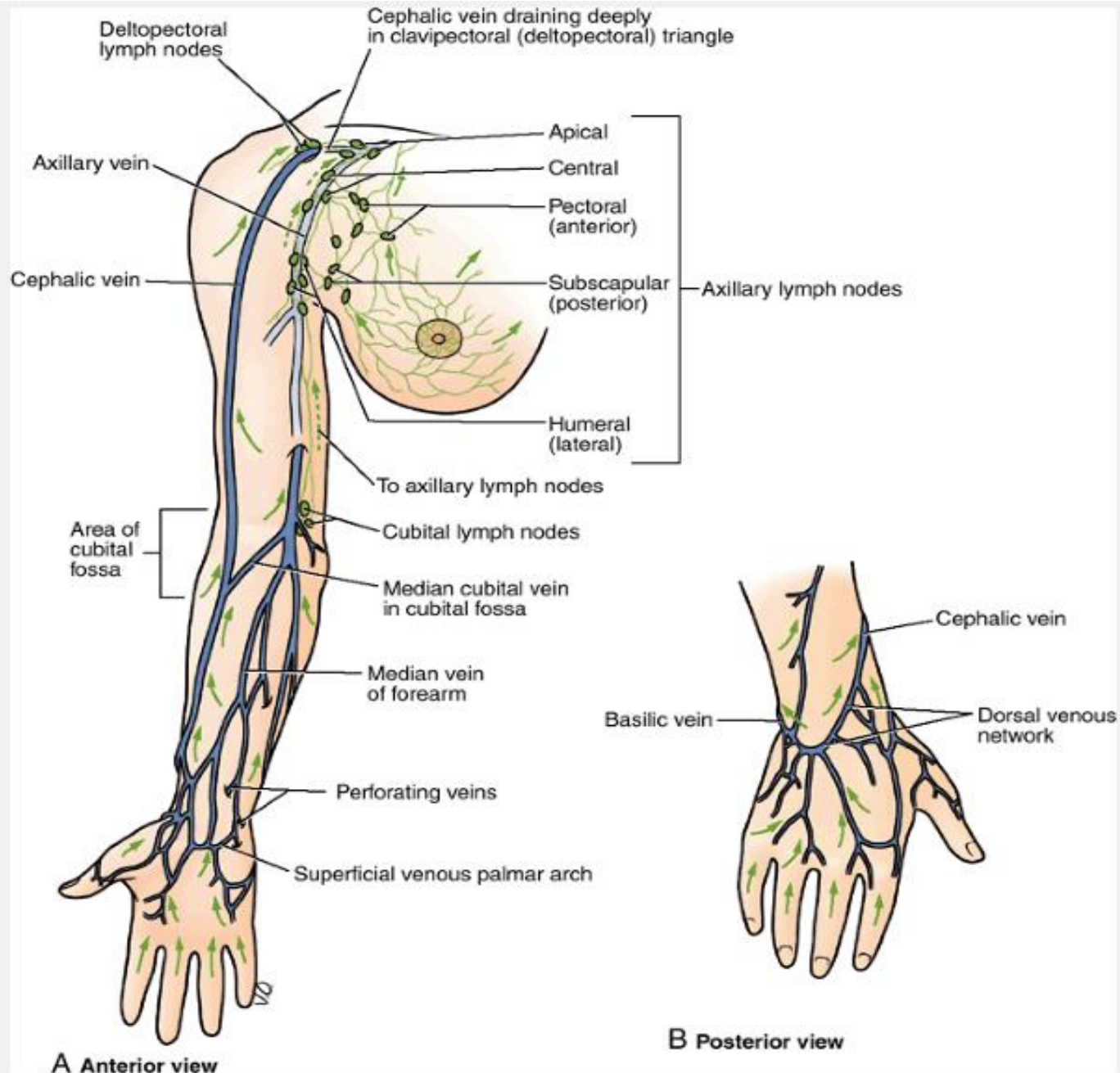
Superficial veins



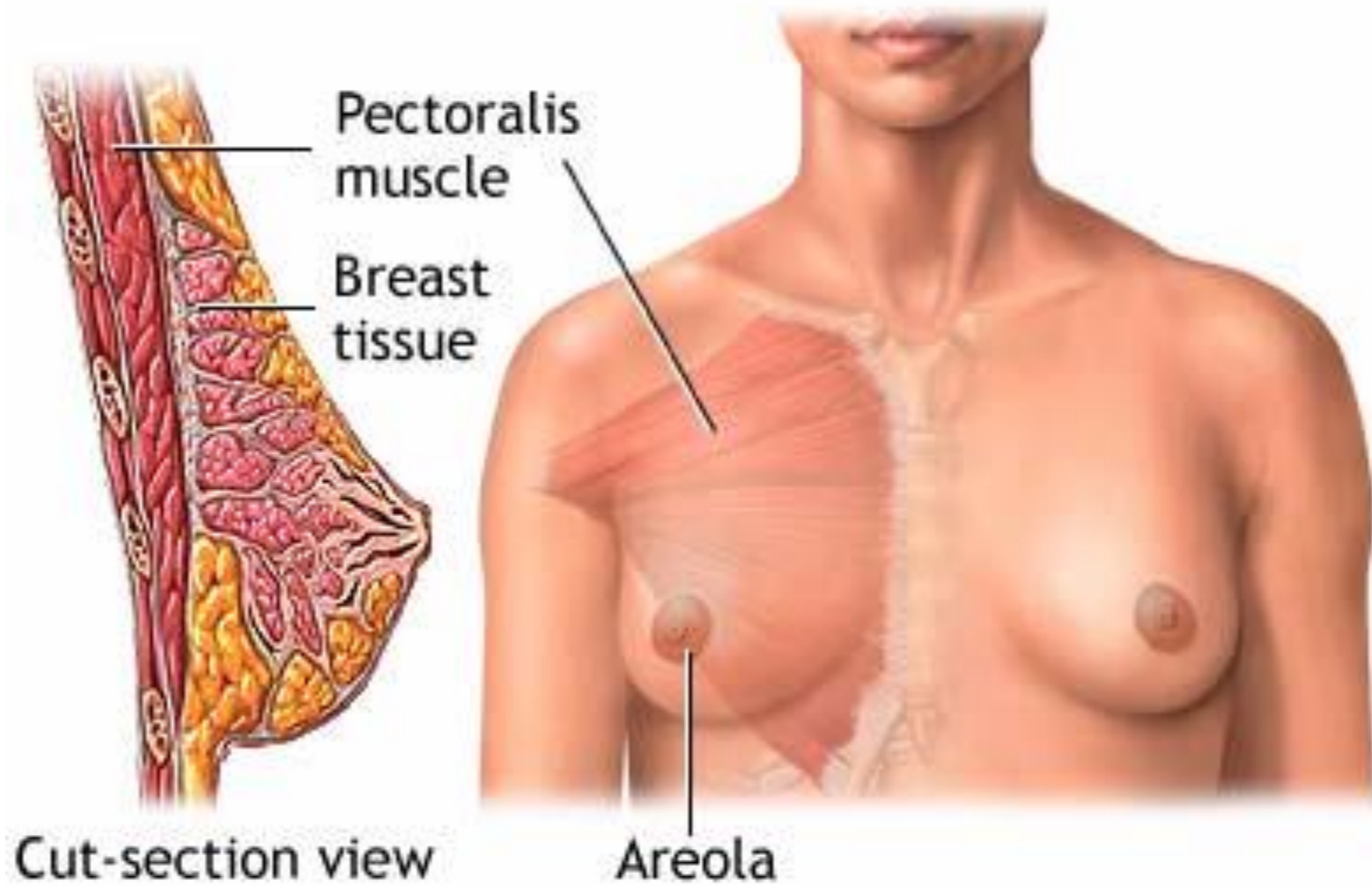
Venipuncture

- Puncture of a vein to draw blood or inject a solution
- Because of the prominence and accessibility of the superficial veins, they are commonly used for venipuncture
- The median cubital vein is commonly used for venipuncture
- The veins forming dorsal venous network and the cephalic and basilic veins arising from it are commonly used for long-term introduction of fluids

Superficial lymph nodes



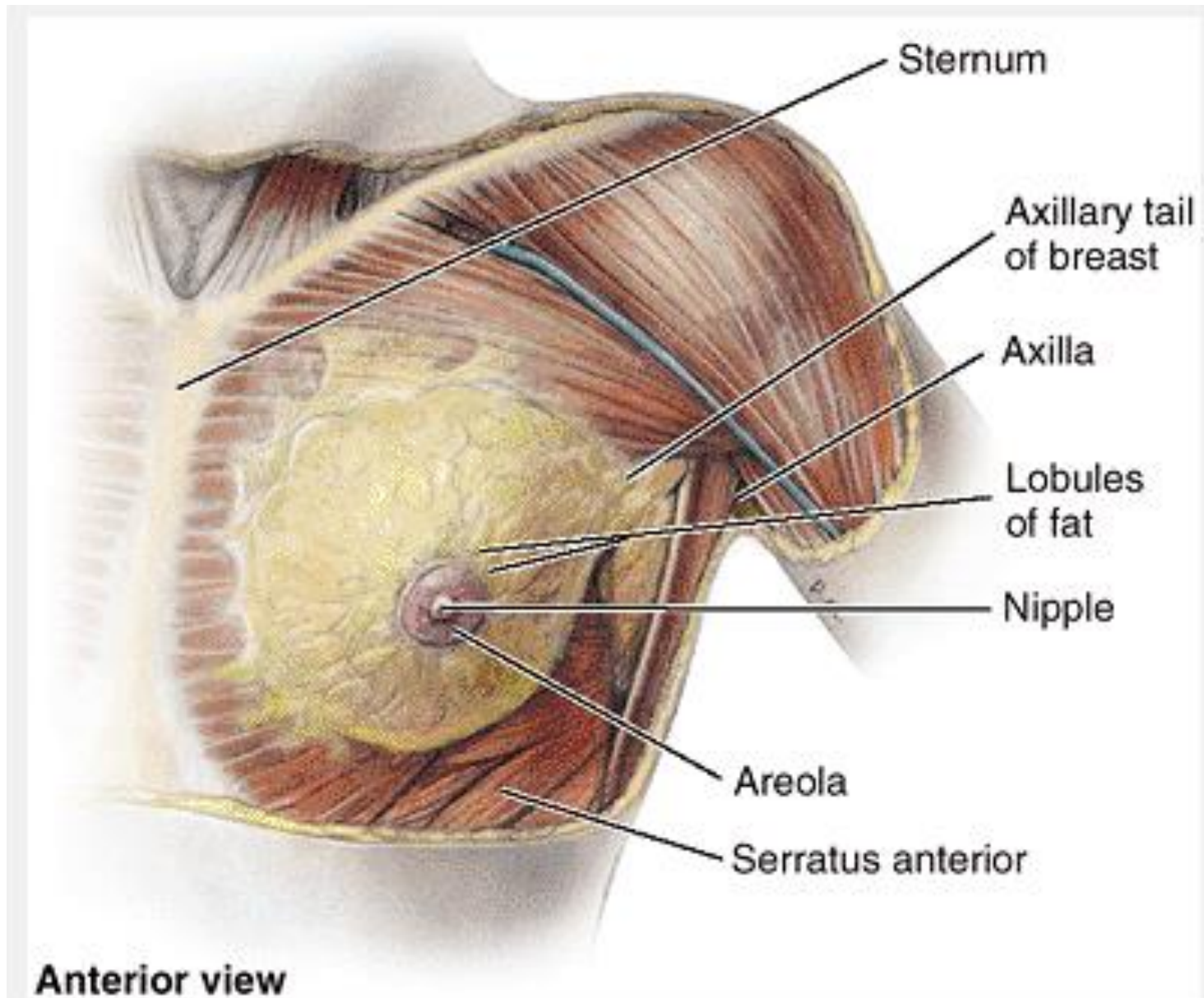
Female breast



Breasts: External anatomy

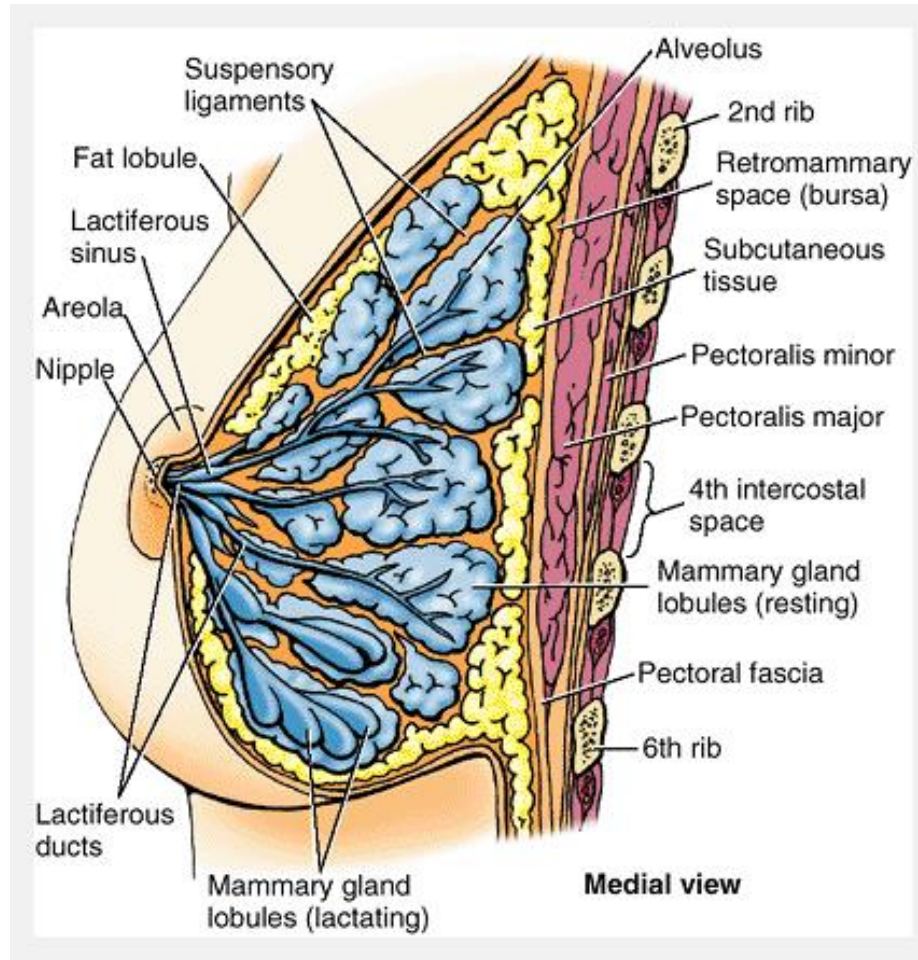
- **Base:** roughly circular; 2nd to 6th ribs and sternum to **midaxillary line**
- Slightly below the center of each breast is a ring of pigmented skin, the **areola**, which surrounds the central conical protruding **nipple**
- Nipple is located at **4th intercostal space** in nulliparous women and male
- A small part of the mammary gland may extend along the inferolateral edge of the pectoralis major toward the axillary fossa (armpit), forming an axillary process or tail

Cont....



Breasts: Internal anatomy

- Lies in **superficial fascia**
- Between breast and deep fascia on pectoral muscle is loose connective tissue plane or potential space - **retromammary space**; allows breast to move freely
- Each mammary gland consists of **15 to 20 lobes** that radiate around and open at the nipple
- The lobes are separated by fat and fibrous connective tissue



Cont....

- The interlobar connective tissue forms **suspensory ligaments** that attach the breast to the underlying muscle fascia and to the overlying skin
- The suspensory ligaments provide natural support for the breasts
- Within the **lobes** are smaller units called **lobules** which contain **glandular alveoli** that produce **milk** when **lactating**
- These compound alveolar glands pass milk into the **lactiferous ducts**, which open to the outside at the nipple
- Just deep to the areola, each lactiferous duct has a dilated region called a **lactiferous sinus**
- Milk accumulates in these sinuses during nursing

Non-lactating and lactating women

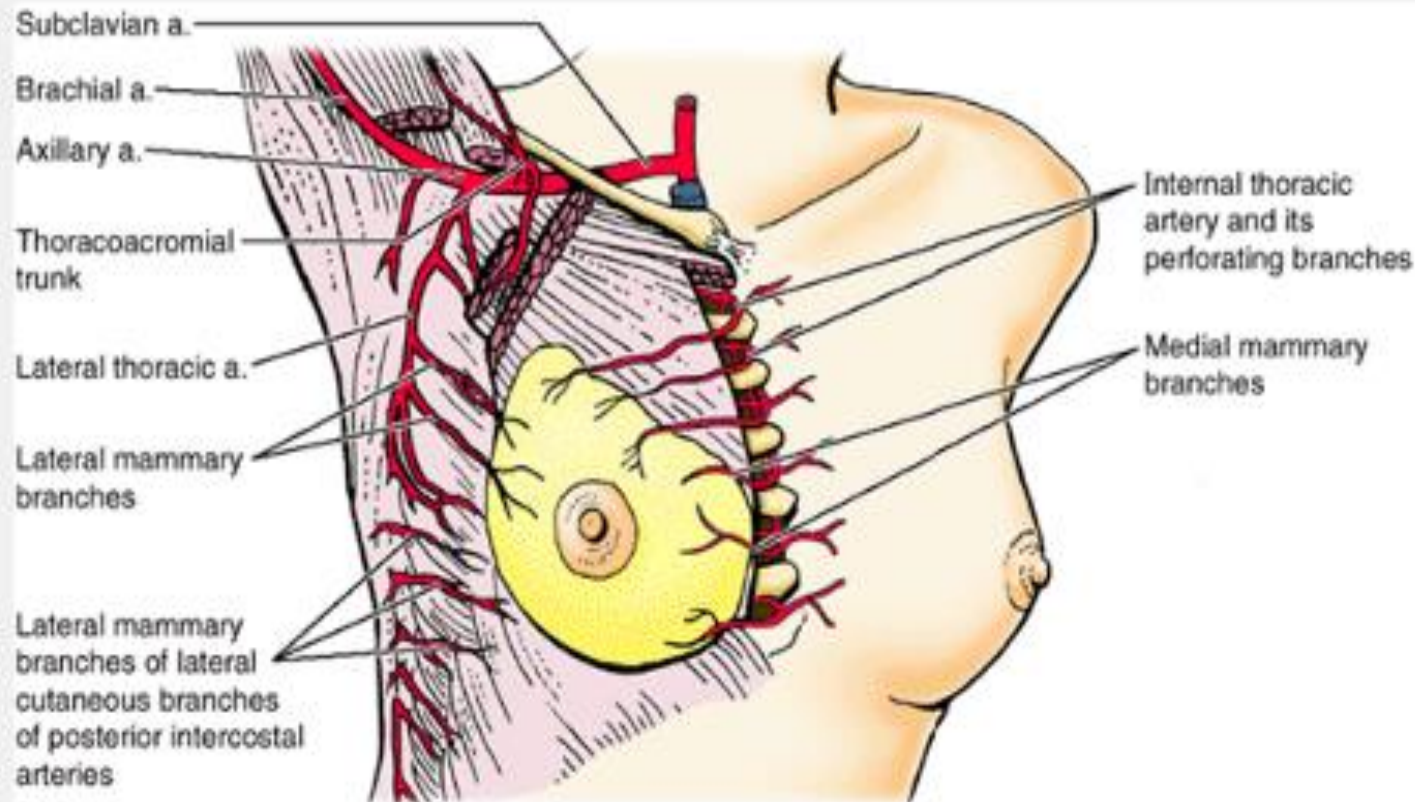
- ❖ Non-lactating women, the predominant component of the breasts is fat, while glandular tissue is more abundant in lactating women
- ❖ A layer of loose connective tissue (the **retromammary space**) separates the breast from the deep fascia and provides some degree of movement over underlying structures
- ❖ The base, or attached surface, of each breast extends vertically from ribs II to VI, and transversely from the sternum to as far laterally as the midaxillary line

Blood supply

Laterally: from **axillary artery**

Medially: from **internal thoracic artery**

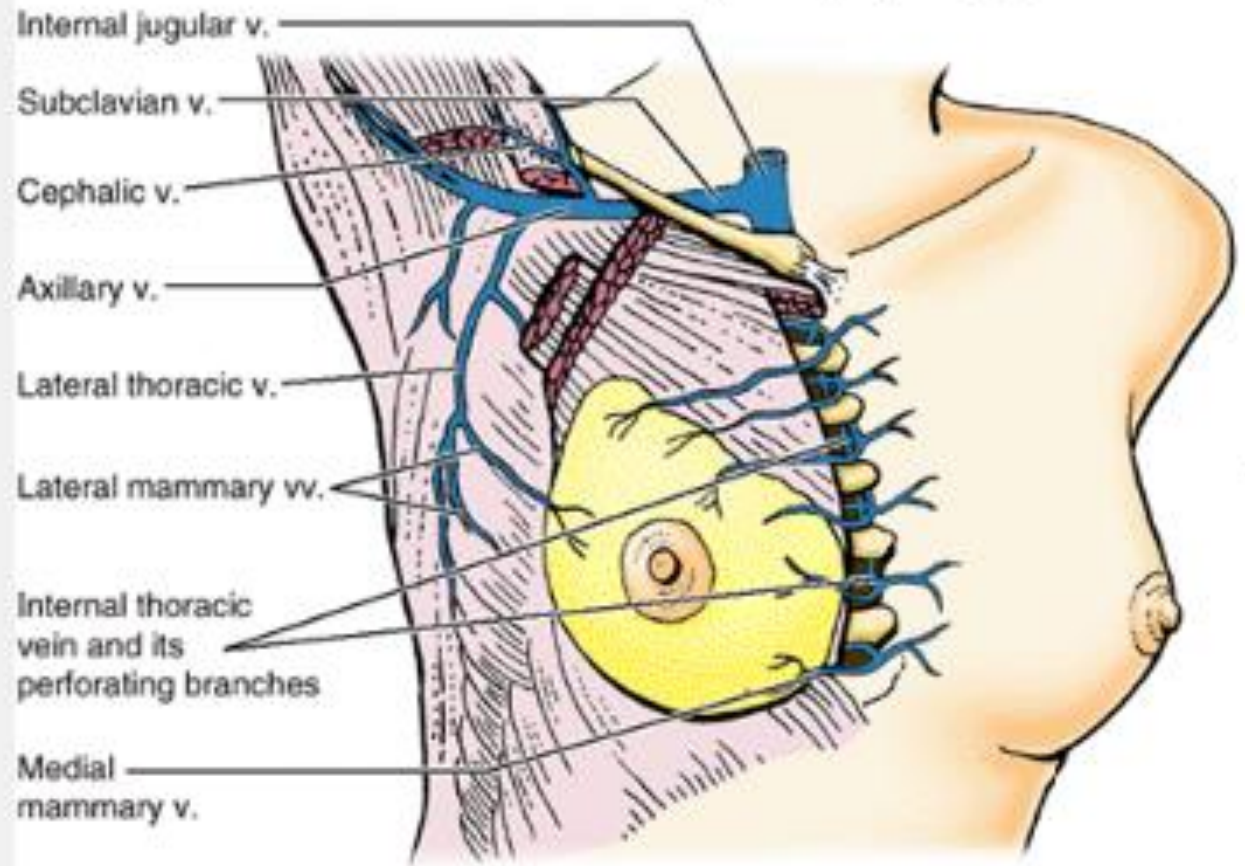
Deep: from 2nd to 4th **intercostal arteries**



(A) Arteries of mammary gland
Anterior (and slightly oblique) view

Venous drainage

- Mainly to the **axillary vein**
- Some drainage to the **internal thoracic vein**



(C) Veins of mammary gland
Anterior (and slightly oblique) view

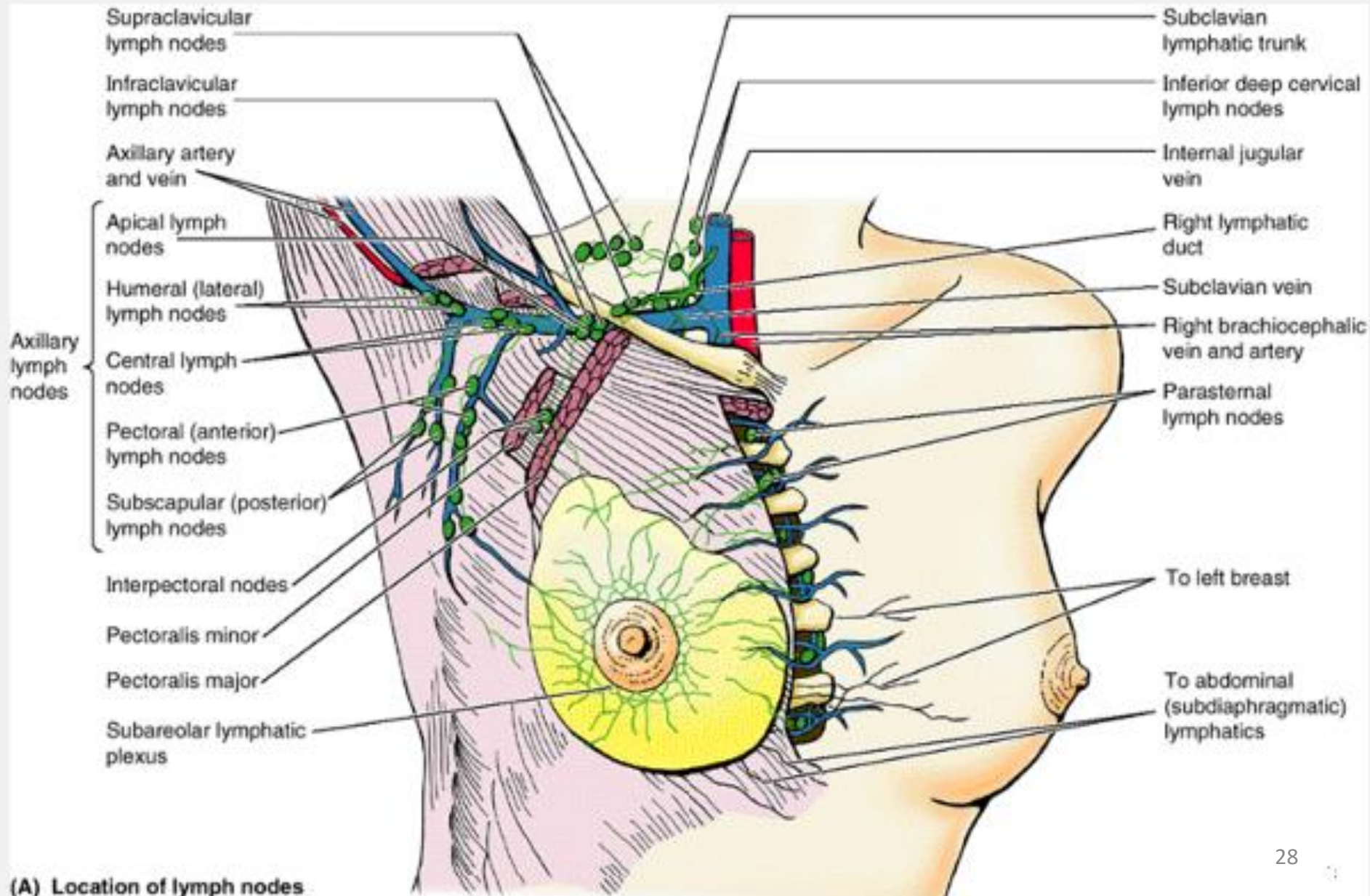
Lymphatic drainage

- Important because of its role in the **metastasis of cancer cells**
- Lymph passes from the **nipple, areola, and lobules of the gland** to the **subareolar lymphatic plexus**
- From this plexus:
 - Most lymph (**> 75%**), especially from the **lateral breast quadrants**, drains to the **axillary lymph nodes**, initially to the **anterior or pectoral nodes** for the most part
 - However, some lymph may drain directly to other axillary nodes or even to interpectoral, deltopectoral, supraclavicular, or inferior deep cervical nodes

Cont....

- Most of the remaining lymph, particularly from the medial breast quadrants, drains to the **parasternal lymph nodes** or to the **opposite breast**, whereas lymph from the **inferior quadrants** may pass deeply to **abdominal lymph nodes**
- Lymph from the skin of the breast, except the nipple and areola, drains into the ipsilateral axillary, inferior deep cervical, and infraclavicular lymph nodes and also into the parasternal lymph nodes of both sides

Cont....



Innervation

- Innervation of the breast is via **anterior and lateral cutaneous branches of the 2nd to 6th intercostal nerves**
- The nipple is innervated by the **4th intercostal nerve**

Clinical correlates

□Breast cancer

- Interference of lymphatic drainage by cancer may cause lymphedema, which results in deviation of nipple and thickening of skin
- Prominent skin between dimpled pores may develop due to involvement of suspensory ligaments
- Because most of lymphatic drainage of the breast is to the axillary lymph nodes, they are the most common site of metastasis from a breast cancer

Breast quadrants

- For the anatomical location and description of tumors and cysts, the surface of the breast is divided into four quadrants

